MITCHELL COHEN

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Robotics PhD candidate specializing in visual-inertial navigation with relevant industry collaboration experience. Strong foundation in estimation theory and practical expertise developing real-time algorithms for autonomous systems. Co-authored 9 peer-reviewed articles in robotics conferences and journals with over 200 citations.

EDUCATION

Doctor of Philosophy in Mechanical Engineering *2020 - 2026 (expected)*
McGill University, Montreal, QC
Advancing Methods for Consistent and Efficient Visual-Inertial Navigation

Master of Mechanical Engineering *2018 - 2020*
McGill University, Montreal, QC
Specialization in dynamics, control, and estimation of aerial autonomous systems. GPA: 3.83/4.00

Bachelor of Mechanical Engineering *2014 - 2018*
McGill University, Montreal, QC
Graduated with distinction. GPA: 3.72/4.00

RELEVANT EXPERIENCE

DECAR Lab, McGill University *2020 - 2026*
PhD Researcher

- Formulated and implemented consistent filtering and optimization-based visual-inertial navigation methods for monocular, stereo, and RGB-D platforms, benchmarking on public and self-collected datasets.
- Designed and implemented point-and-plane-based visual-inertial navigation incorporating loop closures to reduce long-term navigation drift, within an efficient filtering-based framework.
- Performed offline target-based camera-IMU calibration utilizing Kalibr, IMU noise characterization using Allan variance, and online self-calibration of visual-inertial systems.

DENSO Corporation *2020 - 2024*
Research Collaborator

- Researched and developed real-time visual-inertial navigation algorithms for automotive applications.
- Conducted a 3-month internship (Sept. 2023 - Dec. 2023) at the DENSO R&D office in Tokyo, focused on improving sensor calibration and performing simultaneous egomotion estimation and multi-object tracking.
- Delivered technical tutorials to DENSO engineers on visual-inertial navigation methods.

ARA Robotique *2019 - 2020*
Research Intern

- Researched guidance, navigation, and control strategies for unconventional vertical takeoff and landing (VTOL) unmanned aerial vehicle (UAV).

TECHNICAL SKILLS

Languages:	C++, Python, MATLAB
Estimation Theory:	Sensor fusion, SLAM, spatial and temporal sensor calibration, visual-inertial navigation, Lie group theory, nonlinear optimization
Libraries/Frameworks:	Eigen, Ceres, GTSAM, OpenCV, ROS1/ROS2
Tools:	Linux, Git, Docker

PUBLICATIONS

- **M. Cohen**, V. Korotkine, and J. R. Forbes, "Observability and Consistency Analysis for Visual-Inertial Navigation with Anchored Feature Parameterizations", to appear at the *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2026.
- V. Korotkine, **M. Cohen**, and J. R. Forbes, "Globally Optimal Data-Association-Free Landmark-Based Localization Using Semidefinite Relaxations", in *Robotics and Automation Letters (RA-L)*, 2025.
- V. Korotkine, **M. Cohen**, and J. R. Forbes, "A Hessian for Gaussian Mixture Likelihoods in Nonlinear Least Squares", in *Robotics and Automation Letters (RA-L)*, 2024.
- C. C. Cossette, **M. Cohen**, V. Korotkine, A. D. C. Bernal, M. A. Shalaby, and J. R. Forbes, "navlie: A Python Package for State Estimation on Lie Groups", in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023.*
- D. Lisus, **M. Cohen**, and J. R. Forbes, "Know What You Don't Know: Consistency in Sliding Window Filtering with Unobservable States Applied to Visual-Inertial SLAM", *IEEE Robotics and Automation Letters (RA-L)*, 2023.
- **M. Cohen**, and J. R. Forbes "Navigation and Control of Unconventional VTOL UAVs in Forward-Flight with Explicit Wind Velocity Estimation", *Robotics and Automation Letters (RA-L)*, 2020.
- N. van Der Laan, **M. Cohen**, J. Arsenault, and J. R. Forbes, "The Invariant Rauch-Tung-Striebel Smoother", *Robotics and Automation Letters (RA-L)*, 2020.
- **M. Cohen**, K. Abdulrahim, and J. R. Forbes, "Finite-Horizon LQR Control of Quadrotors on $SE_2(3)$", *Robotics and Automation Letters (RA-L)*, 2020.
- R. Chiappinelli, **M. Cohen**, M. Doff-Sotta, M. Nahon, J. R. Forbes, and J. Apkarian, "Modeling and Control of a Passively-Coupled Tilt-Rotor Vertical Takeoff and Landing Aircraft", *International Conference on Robotics and Automation (ICRA)*, 2019.